UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
EASTER 2026 SEMESTER EXAMINATION

PROGRAM: BACHELOR OF SCIENCE IN COMPUTER SCIENCE

YEAR: 3

SEMESTER: 2

COURSE CODE: CSC3218

COURSE NAME: DEEP LEARNING

EXAMINATION TYPE: PROJECT BASED - 100%

EXAMINATION DATE: APRIL 2026

TIME ALLOWED: 2 WEEKS

Examination Instructions

1. The general Uganda Christian University examination guidelines and academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.

2. The examination contains two **COMPULSORY** parts: Part One includes practical deep learning tasks worth 60 marks, and Part Two includes report writing and presentation worth 40 marks.

3. You are required to download the CIFAR-10 dataset. Design a deep learning model that classifies images from the CIFAR-10 dataset, demonstrating the deep learning principles in ConvNet architecture, training strategy, evaluation, and interpretation of results. The following guidance on the required tasks will help you.

4. You are also required to prepare an 8–10 slides PowerPoint presentation that will be delivered in class on the scheduled day of the examination.

5. Submission Requirements: You must submit the following before the scheduled Presentation date of the deep learning examination:

   - Source code
   - 4-page report (PDF)
   - Presentation slides

- Training results and figures

6. Answers to every question should start on a new paper within the examination answer booklet.
7. Uganda Christian University or its officials reserve the right of admission or disqualification from this exam.

PART ONE: TASKS (60 MARKS)

1. **Task 1: Data Preparation (10 marks)**

   You must:

   (a) Load the CIFAR-10 dataset.

   (b) Normalize pixel values.

   (c) Split the dataset into:
   - Training set
   - Validation set
   - Test set

   (d) Perform data augmentation, such as but not limited to:
   - Random cropping
   - Horizontal flipping
   - Rotation (optional)

   Note that you should explain why augmentation is useful.

2. **Task 2: CNN Model Design (20 marks)**

   You must design a Convolutional Neural Network architecture including the following components.

   Minimum requirements:

   - Convolution layers of your choice
   - Activation functions
   - Pooling layers
   - Batch normalization
   - Dropout
   - Fully connected or Global Average Pooling layer
   - Softmax output layer

   You should clearly explain:

   - Why the architecture was chosen
   - The purpose of each layer

3. **Task 3: Model Training (20 marks)**

   You must:

   (a) Train the model using a deep learning framework such as:
   - TensorFlow
   - Keras
   - PyTorch

   (b) Specify training parameters such as:
   - Optimizer (Adam or SGD)

- Learning rate
- Batch size
- Number of epochs
- Etc.

(c) Apply training strategies such as:

- Learning rate scheduling
- Weight decay (regularization)
- Early stopping
- Data augmentation

Please justify why you have or have not used them.

4. **Task 4: Model Evaluation (10 marks)**

   You must evaluate the model using:

   - Training accuracy
   - Validation accuracy
   - Test accuracy, Precision, Recall, and F1 Score

   You must present:

   - Accuracy curves
   - Loss curves
   - Confusion matrix
   - Sample predictions

   You should discuss:

   - Overfitting or underfitting
   - Model performance per class

PART TWO: REPORT AND PRESENTATION (40 MARKS)

1. **Report (4 Pages Maximum) (20 Marks)**

   You must submit a 4-page technical report containing the following sections:

   (a) **Introduction**
   - Brief explanation of image classification
   - Description of the CIFAR-10 dataset

   (b) **Methodology**
   - CNN architecture
   - Training procedure
   - Hyperparameters

   (c) **Results**

   Include but not limited to:

- Training vs validation curves
- Test accuracy, precision, Recall and F1 Score
- Confusion matrix
- Sample predictions

(d) **Discussion**

Discuss among others:

- Model performance
- Overfitting/underfitting
- Possible improvements

(e) **Conclusion**

- Summarize findings and lessons learned.

2. **Presentation (20 marks)**

Students must prepare presentation slides (8–10 slides).

The presentation should include:

(a) Problem statement
(b) Dataset description
(c) Model architecture
(d) Training procedure
(e) Results
(f) Performance analysis
(g) Improvements
(h) Conclusion

Presentation time: 10 minutes.

◀ END ▶

CSC3218 DEEP LEARNING EXAMINATION MARKING GUIDE

Task	Component	Marks	Assessment Criteria
Task 1: Data Preparation	Dataset loading	2	Correct loading of CIFAR-10
	Data normalization	2	Proper pixel normalization
	Dataset splitting	3	Correct split sets
	Data augmentation	2	Implementation of techniques
	Explanation of augmentation	1	Clear justification
Total		**10**	

Task	Component	Marks	Assessment Criteria
Task 2: CNN Model Design	CNN architecture design	6	Logical and structured design
	Required layers implemented	8	all layers
	Explanation of architecture	4	Justification of choices
	Explanation of layer functions	2	Correct layer description
Total		**20**	

Task	Component	Marks	Assessment Criteria
Task 3: Model Training	Framework implementation	5	Correct use of the framework
	Training parameter configuration	5	All hyperparameters
	Training strategies	5	All strategies
	Justification of choices	5	Explanation of training decisions
Total		**20**	

Task	Component	Marks	Assessment Criteria
Task 4: Model Evaluation	Performance metrics	3	Accuracy, precision, recall, F1 score
	Visualization of results	3	Accuracy and loss curves
	Confusion matrix	2	Interpretation of classification results
	Performance discussion	2	Analysis of class results
Total		**10**	

Report Section	Marks	Assessment Criteria
Introduction	3	Explanation of image classification and CIFAR-10
Methodology	5	CNN architecture, training process, hyperparameters
Results	5	Evaluation metrics, curves, and predictions
Discussion	4	Analysis of performance and limitations
Conclusion	3	Summary of findings and lessons learned
Total	**20**	

Presentation Component	**Marks**	**Assessment Criteria**
Problem and dataset explanation	3	Clear explanation of task and data
Model architecture explanation	4	Clear description of CNN structure
Training procedure	3	Explanation of training setup
Results presentation	4	Visualization and explanation of results
Performance analysis	3	Interpretation of model behaviour
Improvements and conclusion	3	Reflection and conclusions
Total	**20**	